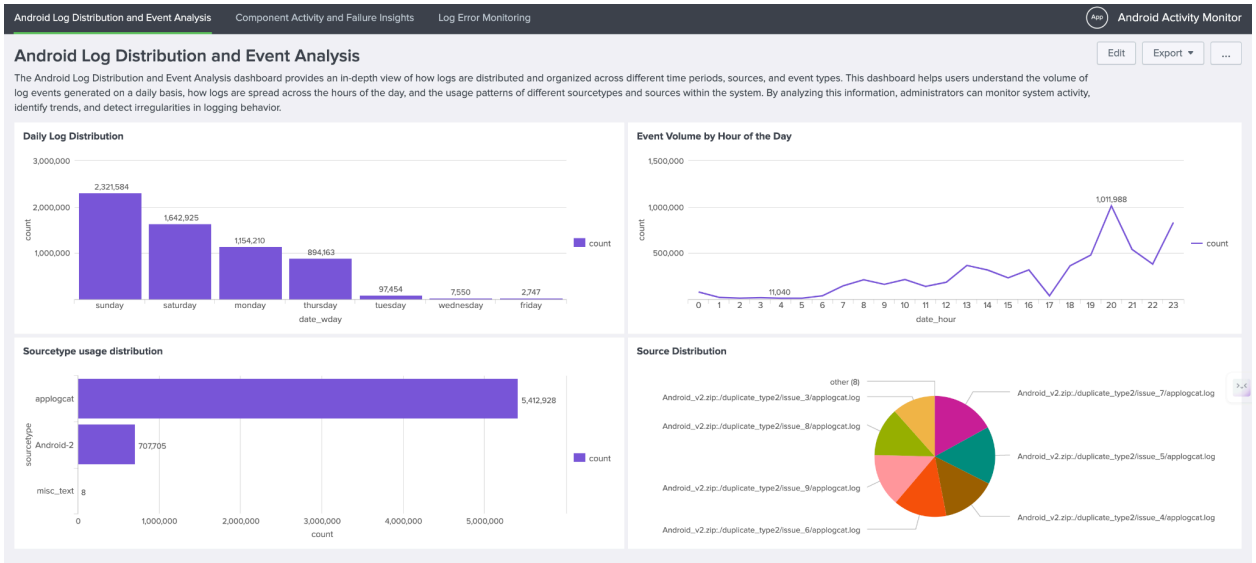
Android Activity Monitor

DASHBOARD 1: Android Log Distribution and Event Analysis



"This dashboard provides a high-level overview of log data generated by the Android system. It helps us understand how logs are distributed across different time periods, sources, and event types."

R1 : Daily Log Distribution

- This chart shows the daily log distribution over a week. We can see that the og volume is consistently higher on weekdays compared to weekends
- This might be related to increased user activity during weekdays.

R2: Event Volume by Hour of the Day

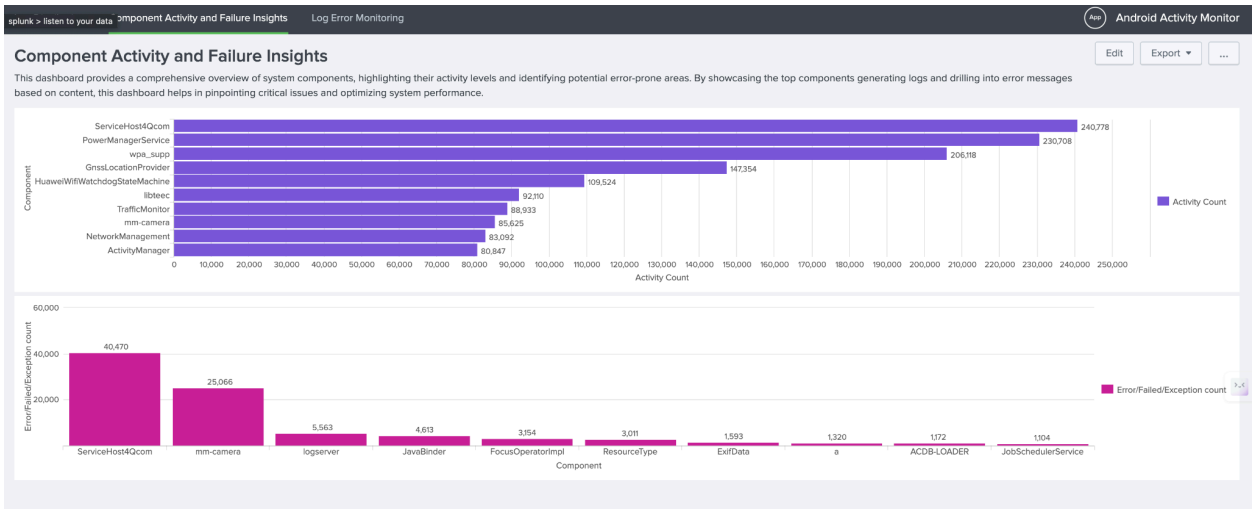
- Here we have the event volume by hour of the day. We can observe that there's a significant peak in event volume between 8 AM and 10 AM"
- "This could be due to users checking their phones first thing in the morning or apps running scheduled tasks."

R3: Source Type and Source Distribution

- "These charts show the distribution of log events by source type and source. The dominant source type is the system logs, which is expected as they capture a wide range of system events"
- "The major contributing sources are the Android framework and network services, indicating that these areas generate a significant portion of the log data"

CONCLUSION : "This dashboard provides a valuable baseline understanding of log generation patterns. By monitoring these trends, we can identify normal behavior, detect anomalies, and gain insights into overall system usage."

DASHBOARD 2: Component Activity and Failure Insights



"This dashboard focuses on the activity levels and error rates of individual components within the Android system. It helps us identify potential bottlenecks, diagnose crashes, and understand component-specific behavior."

R1: Component Activity

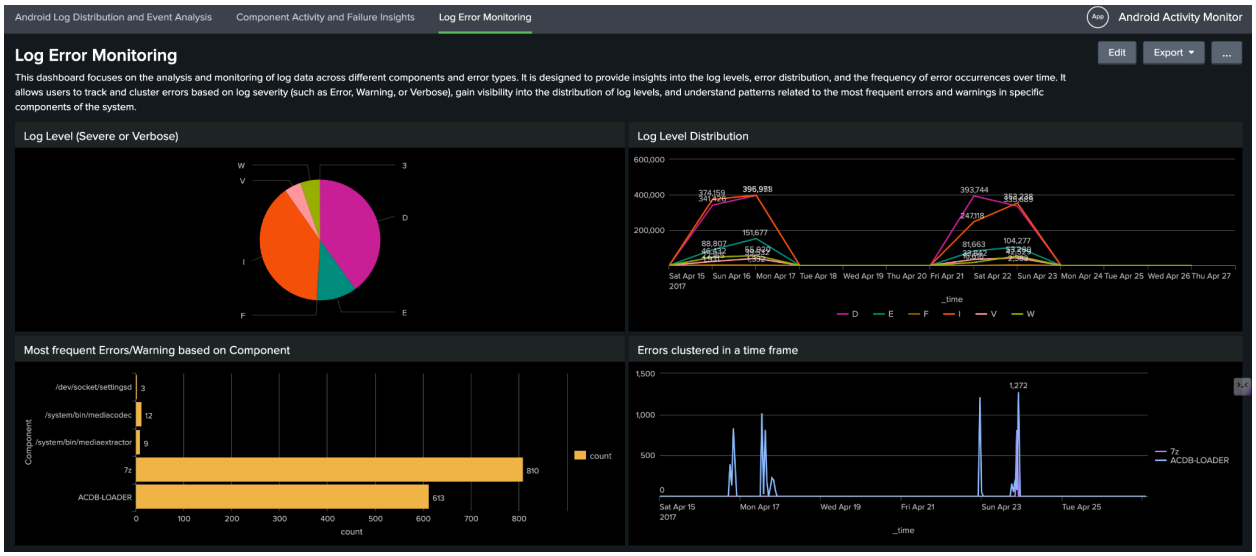
- "This bar chart shows the activity count for various system components. The most active components are [mention the most active components, e.g., ServiceHost4Gcom, PowerManagerService]. This aligns with their roles in managing network connectivity and power management"
- The activity of *activitymanager* is surprisingly low, which might indicate a malfunction.

R2: Error/Failed/Exception Count

- "This chart displays the number of errors, failed operations, and exceptions generated by different components.Component with the highest error count, which is *ServiceHost4Gcom,* are encountering the most issues. This could be attributed to the challenges in maintaining stable network connections"

Conclusion - "By monitoring component-level activity and errors, we can identify potential bottlenecks, diagnose crashes, and prioritize debugging efforts to improve the stability and performance of individual components."

DABOARD 3 : Log Error Monitoring



"This dashboard provides a focused view of error logging across different components. It helps us identify recurring issues, understand error trends, and prioritize areas for improvement within the Android system."

R1: Log Level Between Errors

- "This pie chart shows the distribution of log levels for error messages. The majority of errors are categorized as [dominant level, e.g., Error, Warning], indicating [interpretation, e.g., a significant number of critical errors that require immediate attention]."

R2 : Log Level Distribution and Error Clusters

- "These visualizations show the log level distribution and error clusters over time. We can observe [describe any key trends or patterns, e.g., increase/decrease in certain log levels, spikes at specific times]. This might suggest [potential causes, e.g., software updates, changes in user behavior, external factors like network issues]."

R3 : Most Frequent Errors/Warning based on Component

- "This bar chart highlights the most frequent errors and warnings associated with specific components. The component generating the most error messages is [mention the component and the number of errors]. This indicates a need for further investigation into [specific areas related to the component, e.g., resource usage or error handling mechanisms]."

Conclusion - "This dashboard provides valuable insights into the types, frequency, and distribution of errors, helping us identify recurring issues, prioritize debugging efforts, and improve the overall stability of the system by addressing the root causes of errors."

OVERALL CONCLUSION :

Option 1 (Concise and impactful):
"In conclusion, these three dashboards demonstrate the power of Splunk for analyzing Android log data. By providing insights into log patterns, component behavior, and error trends, we can proactively address issues, optimize performance, and improve the overall stability and reliability of the Android system."

Option 2 (Highlighting key takeaways):
"To wrap up, we've seen how these dashboards can help us:

- *Understand system-wide log patterns and identify potential anomalies.*
- *Monitor component activity and diagnose issues affecting specific parts of the system.*
- *Analyze error trends to prioritize debugging efforts and improve system stability.*

Ultimately, these insights enable us to make data-driven decisions to enhance the performance and user experience of the Android system."